

Exercises and Solutions

Theory of Complex analysis

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Problem 1.

(i) Use Cauchy's integral formula to compute

$$\int_{|z|=1} \frac{e^{\alpha z}}{2z^2 - 5z + 2} dz, \quad \alpha \in \mathbb{C}.$$

(ii) By considering the real part of a suitable complex integral, show that for $r \in (0, 1)$,

$$\int_0^\pi \frac{\cos(n\theta)}{1 - 2r \cos \theta + r^2} d\theta = \frac{\pi r^n}{1 - r^2}, \quad \int_0^{2\pi} \cos(\cos \theta) \cosh(\sin \theta) d\theta = 2\pi.$$

Background. Cauchy's integral formula: if f is holomorphic on and inside a simple closed curve γ , then $\int_\gamma \frac{f(z)}{z-a} dz = 2\pi i f(a)$ for a inside γ . Also, with $z = e^{i\theta}$ one has $d\theta = \frac{dz}{iz}$ and $1 - 2r \cos \theta + r^2 = (1 - rz)(1 - rz^{-1}) = \frac{(z-r)(1-rz)}{z}$.

Solution. (i) Factor $2z^2 - 5z + 2 = 2(z-2)(z-\frac{1}{2})$. On $|z|=1$ only $z = \frac{1}{2}$ lies inside. Hence

$$\int_{|z|=1} \frac{e^{\alpha z}}{2z^2 - 5z + 2} dz = 2\pi i \cdot \text{Res} \left(\frac{e^{\alpha z}}{2(z-2)(z-\frac{1}{2})}; \frac{1}{2} \right) = 2\pi i \cdot \frac{e^{\alpha/2}}{2(\frac{1}{2}-2)} = -\frac{2\pi i}{3} e^{\alpha/2}.$$

(ii-a) Let

$$I_n = \int_0^{2\pi} \frac{e^{in\theta}}{1 - 2r \cos \theta + r^2} d\theta.$$

With $z = e^{i\theta}$ and $d\theta = \frac{dz}{iz}$,

$$I_n = \oint_{|z|=1} \frac{z^n}{(z-r)(1-rz)} \frac{dz}{i}.$$

Only $z = r$ lies inside, giving $\text{Res} = \frac{r^n}{1-r^2}$. Thus

$$I_n = 2\pi \frac{r^n}{1-r^2}, \quad \int_0^{2\pi} \frac{\cos(n\theta)}{1 - 2r \cos \theta + r^2} d\theta = \Re I_n = \frac{2\pi r^n}{1-r^2}.$$

Halving the interval,

$$\int_0^\pi \frac{\cos(n\theta)}{1 - 2r \cos \theta + r^2} d\theta = \frac{\pi r^n}{1-r^2}.$$

(ii-b) Using $\cos(\cos \theta) \cosh(\sin \theta) = \frac{1}{2}(e^{ie^{i\theta}} + e^{-ie^{i\theta}})$,

$$\int_0^{2\pi} \cos(\cos \theta) \cosh(\sin \theta) d\theta = \frac{1}{2} \oint_{|z|=1} \frac{e^{iz} + e^{-iz}}{iz} dz = \frac{1}{2i} \cdot 2\pi i (e^0 + e^0) = 2\pi.$$

□

Residues: definitions, equivalences, and tools. *Laurent series and the residue.* Let f be meromorphic with an isolated singularity at a . Then there is $\rho > 0$ and integers $m \geq 0$ and coefficients $\{c_k\}_{k=-m}^{\infty}$ such that

$$f(z) = \sum_{k=-m}^{\infty} c_k (z-a)^k, \quad 0 < |z-a| < \rho.$$

Definition. The *residue* of f at a is the coefficient of $(z-a)^{-1}$:

$$\boxed{\text{Res}(f; a) = c_{-1}}.$$

Equivalent characterizations. For any $0 < \varepsilon < \rho$,

$$\text{Res}(f; a) = \frac{1}{2\pi i} \oint_{|z-a|=\varepsilon} f(z) dz.$$

If a is a simple pole (i.e. $m = 1$), then

$$\text{Res}(f; a) = \lim_{z \rightarrow a} (z-a) f(z).$$

More generally, if a is a pole of order m ,

$$\text{Res}(f; a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} ((z-a)^m f(z)).$$

Proof sketch of c_{-1} -rule. Write the Laurent series and integrate termwise on $|z-a| = \varepsilon$:

$$\oint (z-a)^k dz = \begin{cases} 0, & k \neq -1, \\ 2\pi i, & k = -1. \end{cases}$$

Thus $\oint f = 2\pi i c_{-1}$ and $\text{Res}(f; a) = c_{-1}$.

Residue Theorem (workhorse). If f is meromorphic on and inside a positively oriented simple closed curve γ with no poles on γ , then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{a_k \in \text{Int}(\gamma)} \text{Res}(f; a_k).$$

Quick computation rules.

- **Simple pole formula.** If $f(z) = \frac{p(z)}{q(z)}$ with p, q holomorphic, $q(a) = 0$, $q'(a) \neq 0$, then

$$\text{Res}(f; a) = \frac{p(a)}{q'(a)}.$$

- **Analytic factor.** If g is holomorphic near a , then $\text{Res}(g f; a) = g(a) \text{Res}(f; a)$.
- **Pole at infinity.** If f is meromorphic on the Riemann sphere $\widehat{\mathbb{C}}$,

$$\text{Res}(f; \infty) := - \sum_{a \in \mathbb{C}} \text{Res}(f; a), \quad \oint_{|z|=R} f(z) dz \xrightarrow{R \rightarrow \infty} -2\pi i \text{Res}(f; \infty),$$

when the outer integral stabilizes.

Micro-examples (via c_{-1}).

- $f(z) = \frac{1}{(z-a)(z-b)}$ near a : expand $\frac{1}{z-b} = \frac{1}{a-b} + O(z-a)$, so

$$f(z) = \frac{1}{z-a} \cdot \frac{1}{a-b} + O(1) \Rightarrow \text{Res}(f; a) = \frac{1}{a-b}.$$

- $f(z) = \frac{e^{\alpha z}}{(z-a)(z-b)}$ near a : $e^{\alpha z} = e^{\alpha a} + O(z-a)$, hence

$$f(z) = \frac{1}{z-a} \cdot \frac{e^{\alpha a}}{a-b} + O(1) \Rightarrow \text{Res}(f; a) = \frac{e^{\alpha a}}{a-b}.$$

Unit-circle substitution for trig integrals. With $z = e^{i\theta}$, $d\theta = \frac{dz}{iz}$ and

$$1 - 2r \cos \theta + r^2 = (1 - rz)(1 - rz^{-1}) = \frac{(z-r)(1-rz)}{z}.$$

Only $z = r$ lies inside $|z| = 1$ if $0 < r < 1$, giving

$$\int_0^{2\pi} \frac{e^{in\theta}}{1 - 2r \cos \theta + r^2} d\theta = \oint_{|z|=1} \frac{z^n}{(z-r)(1-rz)} \frac{dz}{i} = 2\pi \frac{r^n}{1-r^2}.$$

Taking real parts yields $\int_0^\pi \frac{\cos(n\theta)}{1 - 2r \cos \theta + r^2} d\theta = \frac{\pi r^n}{1-r^2}$.

Problem 2. Find the Laurent expansion (in powers of z , i.e. about 0) of

$$\frac{1}{z^2 - 3z + 2} = \frac{1}{(z-1)(z-2)}$$

in the regions $\{|z| < 1\}$, $\{1 < |z| < 2\}$, and $\{|z| \geq 2\}$.

Background. Partial fractions:

$$\frac{1}{(z-1)(z-2)} = -\frac{1}{z-1} + \frac{1}{z-2}.$$

Geometric expansions:

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n \quad (|w| < 1), \quad \frac{1}{z-a} = \frac{1}{z} \cdot \frac{1}{1-\frac{a}{z}} = \sum_{n=0}^{\infty} \frac{a^n}{z^{n+1}} \quad (|z| > |a|).$$

Solutions. (A) For $|z| < 1$.

$$-\frac{1}{z-1} = \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \quad \frac{1}{z-2} = -\frac{1}{2} \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}.$$

Hence

$$\boxed{\frac{1}{z^2-3z+2} = \sum_{n=0}^{\infty} (1-2^{-(n+1)})z^n, \quad |z| < 1.}$$

(B) For $1 < |z| < 2$.

$$-\frac{1}{z-1} = -\frac{1}{z} \frac{1}{1-\frac{1}{z}} = -\sum_{n=1}^{\infty} z^{-n}, \quad \frac{1}{z-2} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}.$$

Thus

$$\boxed{\frac{1}{z^2-3z+2} = -\sum_{n=1}^{\infty} z^{-n} - \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \quad 1 < |z| < 2.}$$

(C) For $|z| > 2$.

$$-\frac{1}{z-1} = -\sum_{m=1}^{\infty} z^{-m}, \quad \frac{1}{z-2} = \frac{1}{z} \frac{1}{1-\frac{2}{z}} = \sum_{m=1}^{\infty} \frac{2^{m-1}}{z^m}.$$

Therefore

$$\boxed{\frac{1}{z^2-3z+2} = \sum_{m=1}^{\infty} (2^{m-1}-1)z^{-m}, \quad |z| > 2.}$$

□

Problem 3. Classify the singularities of

$$\frac{z}{\sin z}, \quad \sin\left(\frac{\pi}{z^2}\right), \quad \frac{1}{z^2} + \frac{1}{z^2+1}, \quad \frac{1}{z^2} \cos\left(\frac{\pi z}{z+1}\right).$$

Background. At an isolated singularity a , if the Laurent expansion $f(z) = \sum_{k=-m}^{\infty} c_k(z-a)^k$ has (i) $m=0 \Rightarrow$ removable; (ii) finitely many negative terms with highest $(z-a)^{-m} \Rightarrow$ pole of order m ; (iii) infinitely many negative terms $\Rightarrow$ essential.

(1) $z/\sin z$. Near 0, $\sin z = z - \frac{z^3}{6} + O(z^5)$, so

$$\frac{z}{\sin z} = 1 + \frac{z^2}{6} + O(z^4),$$

hence $z=0$ is **removable** (define value 1). At $z=n\pi$, $n \in \mathbb{Z} \setminus \{0\}$, $\sin z$ has simple zeros and the numerator is nonzero, so each $n\pi$ is a **simple pole**.

(2) $\sin(\pi/z^2)$. Using $\sin w = \sum_{k \geq 0} \frac{(-1)^k w^{2k+1}}{(2k+1)!}$ with $w = \pi/z^2$,

$$\sin\left(\frac{\pi}{z^2}\right) = \sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k+1}}{(2k+1)!} z^{-4k-2},$$

which contains infinitely many negative powers. Thus $z = 0$ is an **essential singularity**. There are no other singularities.

(3) $1/z^2 + 1/(z^2 + 1)$. At $z = 0$ one has a **pole of order 2**. At $z = \pm i$ (simple zeros of $z^2 + 1$) there are **simple poles**. No other singularities occur.

(4) $z^{-2} \cos(\pi z/(z + 1))$. Let $h(z) = \frac{\pi z}{z + 1}$. Near $z = 0$, $h(z) = \pi(z - z^2 + z^3 - \dots)$, hence

$$\cos h(z) = 1 - \frac{1}{2}h(z)^2 + O(z^3) = 1 - \frac{\pi^2}{2}z^2 + O(z^3),$$

and

$$\frac{1}{z^2} \cos h(z) = \frac{1}{z^2} - \frac{\pi^2}{2} + O(z),$$

so $z = 0$ is a **pole of order 2**. At $z = -1$, h has a simple pole; since $\cos w = \frac{1}{2}(e^{iw} + e^{-iw})$, composition with a pole yields an **essential singularity** at $z = -1$. There are no other singularities. $\square$

Problem 4. Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be entire. Prove that f is constant if any one of the following holds:

- (i) $\frac{f(z)}{z} \rightarrow 0$ as $|z| \rightarrow \infty$.
- (ii) There exist $b \in \mathbb{C}$ and $\varepsilon > 0$ such that $|f(z) - b| > \varepsilon$ for all $z \in \mathbb{C}$.
- (iii) Writing $f = u + iv$ with real u, v , one has $|u(z)| > |v(z)|$ for all $z \in \mathbb{C}$.

Background. We use: (a) *Liouville's theorem*: a bounded entire function is constant; (b) *Cauchy estimates*: if f is holomorphic on $|z| \leq R$, then $|f^{(n)}(0)| \leq \frac{n!}{R^n} \max_{|z|=R} |f(z)|$; (c) the Cayley map $w \mapsto \frac{w-1}{w+1}$ sends $\{\Re w > 0\}$ to the unit disk; (d) if $FG \equiv 0$ with F, G holomorphic on a connected set, then $F \equiv 0$ or $G \equiv 0$.

Solution. (i) *Sublinear growth.* Given $\eta > 0$, choose R_η so that $|f(z)| \leq \eta|z|$ for $|z| \geq R_\eta$. For $R \geq R_\eta$,

$$\max_{|z|=R} |f(z)| \leq \eta R.$$

Cauchy estimates at 0 yield

$$|f^{(n)}(0)| \leq \frac{n! \eta R}{R^n} = \frac{n! \eta}{R^{n-1}} \quad (n \geq 1).$$

Letting $R \rightarrow \infty$ gives $f^{(n)}(0) = 0$ for $n \geq 2$, and $|f'(0)| \leq \eta$ for all $\eta > 0$, hence $f'(0) = 0$. Thus f is constant.

(ii) *Image avoids a disk.* If $|f(z) - b| > \varepsilon$ for all z , then $g(z) = \frac{1}{f(z) - b}$ is entire and bounded by $1/\varepsilon$. By Liouville, g is constant, hence f is constant.

(iii) *Real part dominates.* If $f = u + iv$ with $|u| > |v|$ everywhere, then $\Re(f^2) = u^2 - v^2 > 0$, so $g := f^2$ maps $\mathbb{C}$ into the right half-plane. The Cayley transform $h = (g - 1)/(g + 1)$ is entire and satisfies $|h| < 1$, hence h is constant by Liouville, so $g \equiv c$ with $\Re c > 0$. Therefore $(f - \sqrt{c})(f + \sqrt{c}) \equiv 0$, and by analyticity one factor vanishes identically. Thus $f \equiv \sqrt{c}$ or $f \equiv -\sqrt{c}$, i.e. f is constant. $\square$

Problem 6.

- (i) Let f be entire. Show that f is a polynomial of degree $\leq k$ iff there is $M > 0$ with

$$|f(z)| \leq M(1 + |z|)^k \quad (\forall z \in \mathbb{C}).$$

- (ii) Show that an entire f is a polynomial of positive degree iff $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$.
 (iii) Let f be analytic on $\mathbb{C}$ apart from finitely many poles. If there exists k such that $|f(z)| \leq |z|^k$ for all sufficiently large $|z|$, prove that f is a rational function.

Background. *Cauchy estimates.* If f is holomorphic on $|z| \leq R$, then

$$|f^{(n)}(0)| \leq \frac{n!}{R^n} \max_{|z|=R} |f(z)|.$$

Poles at infinity. For $g(w) = f(1/w)$ near $w = 0$, a pole of order m at 0 means f has a pole at ∞ of order m . An entire function with a pole at ∞ is a polynomial (degree m). A function meromorphic on the Riemann sphere (only poles, finitely many) is rational.

Solution. (i) $\Leftarrow$. Assume $|f(z)| \leq M(1 + |z|)^k$. For $R > 0$, $\max_{|z|=R} |f(z)| \leq M(1 + R)^k$. Hence

$$|f^{(n)}(0)| \leq \frac{n!}{R^n} M(1 + R)^k \quad (n \geq 0).$$

Fix $n > k$ and let $R \rightarrow \infty$: $(1 + R)^k/R^n \rightarrow 0$, so $f^{(n)}(0) = 0$. Therefore the Taylor series truncates at degree $\leq k$ and f is a polynomial.

(i) $\Rightarrow$. If $f(z) = \sum_{j=0}^k a_j z^j$, then $|f(z)| \leq \sum_{j=0}^k |a_j| |z|^j \leq C(1 + |z|)^k$.

(ii) $\Rightarrow$. If f is a nonconstant polynomial of degree $m \geq 1$, then $|f(z)| \sim |a_m| |z|^m \rightarrow \infty$.

(ii) $\Leftarrow$. If $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, then $g(w) = f(1/w)$ has a pole at $w = 0$, so f has a pole at ∞ and is therefore a polynomial of positive degree.

(iii). Let $g(w) = f(1/w)$. For small $|w|$,

$$|g(w)| = |f(1/w)| \leq |1/w|^k = |w|^{-k},$$

so g has at most a pole of order $\leq k$ at 0. Hence f has at most a pole at ∞ . Together with the finitely many poles in $\mathbb{C}$, f is meromorphic on the Riemann sphere with only poles; thus f is rational. $\square$

Problem 7.

- (i) (**Schwarz's Lemma**) Let f be analytic on $D(0, 1)$ with $|f(z)| \leq 1$ and $f(0) = 0$. Apply the maximum principle to $f(z)/z$ to show $|f(z)| \leq |z|$. Show also that if $|f(w)| = |w|$ for some $w \neq 0$, then $f(z) = cz$ for some constant c .
 (ii) Use Schwarz's Lemma to prove that any conformal equivalence $\Phi : D(0, 1) \rightarrow D(0, 1)$ is given by a Möbius transformation.

Background. If a holomorphic g on $0 < |z| < 1$ is bounded near 0, then g extends holomorphically to $z = 0$ (removable singularity). A non-constant holomorphic function does not attain a maximum of its modulus in the interior (maximum modulus principle). For $a \in D$, the map $\phi_a(z) = \frac{z - a}{1 - \bar{a}z}$ is a biholomorphism $D \rightarrow D$ with $\phi_a(a) = 0$.

Solution. (i) Define

$$g(z) = \begin{cases} \frac{f(z)}{z}, & z \neq 0, \\ f'(0), & z = 0. \end{cases}$$

Because $|f(z)| \leq 1$ and $f(0) = 0$, g is bounded near 0, hence extends holomorphically to D . For $0 < r < 1$, on $|z| = r$ we have $|g(z)| \leq 1/r$. By the maximum modulus principle on each smaller disk, letting $r \uparrow 1$ gives $|g(z)| \leq 1$ for all $z \in D$. Thus $|f(z)| \leq |z|$ and $|f'(0)| = |g(0)| \leq 1$.

If $|f(w)| = |w|$ for some $w \neq 0$, then $|g(w)| = 1$. By the maximum modulus principle g has constant modulus 1 on D , hence $g \equiv c$ with $|c| = 1$ and $f(z) = cz$.

(ii) Let $\Phi : D \rightarrow D$ be a conformal bijection and set $a = \Phi(0)$. Consider $F = \phi_a \circ \Phi$, where $\phi_a(z) = \frac{z-a}{1-\bar{a}z}$. Then $F : D \rightarrow D$ is holomorphic with $F(0) = 0$. By Schwarz's Lemma, $F(z) = \lambda z$ with $|\lambda| \leq 1$; since Φ is bijective, $|\lambda| = 1$. Therefore

$$\Phi(z) = \phi_a^{-1}(\lambda z) = \lambda \frac{z-a}{1-\bar{a}z},$$

a Möbius automorphism of the disk. □

Conformal maps and conformal equivalence. *Conformal at a point.* A map $f : U \rightarrow \mathbb{C}$ is conformal at $z_0 \in U$ if it preserves angles between smooth curves through z_0 (with orientation). In complex analysis this is equivalent to

$$f \text{ holomorphic at } z_0 \quad \text{and} \quad f'(z_0) \neq 0.$$

Indeed, the first-order expansion $f(z) = f(z_0) + f'(z_0)(z - z_0) + o(|z - z_0|)$ shows f is locally “rotation + dilation”. Writing $f = u + iv$, the Cauchy–Riemann equations at z_0 give

$$Df(z_0) = \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = |f'(z_0)| \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

so $Df(z_0)$ is a positive scalar times a rotation, hence angle-preserving and orientation-preserving.

Angle preservation formula. If $v_1, v_2 \in \mathbb{C}$ are tangent directions at z_0 , then under f

$$\cos \angle(f_*v_1, f_*v_2) = \frac{\Re((f'(z_0)v_1)\overline{(f'(z_0)v_2)})}{|f'(z_0)v_1||f'(z_0)v_2|} = \frac{\Re(v_1\overline{v_2})}{|v_1||v_2|},$$

so the angle is unchanged.

Conformal equivalence. Domains $U, V \subset \mathbb{C}$ are *conformally equivalent* if there is a biholomorphism $f : U \rightarrow V$ (holomorphic and bijective with holomorphic inverse). Such an f is conformal on U , and f^{-1} is conformal on V .

Examples.

- Cayley transform $\phi(z) = \frac{z-i}{z+i}$: biholomorphism $\{\Im z > 0\} \leftrightarrow \mathbb{D}$.
- Exponential $w = e^{\pi z}$: biholomorphism $\{0 < \Im z < 1\} \leftrightarrow \{1 < |w| < e^\pi\}$.

Non-examples.

- $f(z) = z^n$ is not conformal at $z = 0$ for $n \geq 2$ since $f'(0) = 0$ (angles multiply by n).
- Conjugation $f(z) = \bar{z}$ reverses orientation (anti-conformal).

Problem 8 (restated).

- (i) Let f be entire and satisfy $f(1/n) = 1/n$ for all $n \in \mathbb{N}$. Prove that $f(z) \equiv z$.
- (ii) Let f be entire and satisfy $f(n) = n^2$ for all $n \in \mathbb{Z}$. Must f equal z^2 ? Give a justification or a counterexample.
- (iii) Let f be holomorphic on $D(0, 2)$. Show that there exists $n \in \mathbb{N}$ with $f(1/n) \neq 1/(n+1)$.

Background and definitions for Problem 8. *Holomorphic / entire.* A function f is holomorphic on a domain $U \subset \mathbb{C}$ if it is complex differentiable at every point of U . It is *entire* if it is holomorphic on all of $\mathbb{C}$. A *domain* is a nonempty, open, connected subset of $\mathbb{C}$.

Accumulation point. A sequence $(z_n) \subset U$ has an accumulation point $z_* \in U$ if $z_n \rightarrow z_*$ with $z_n \neq z_*$ eventually.

Isolated zeros and the identity theorem. If f is holomorphic on U and not identically zero, then its zeros have no accumulation point in U . If f, g are holomorphic on U and $f = g$ on a set with a limit point in U (e.g. $z_n \rightarrow z_* \in U$), then $f \equiv g$ on U . Equivalently, if a holomorphic f vanishes on such a set, then $f \equiv 0$.

An entire function vanishing on $\mathbb{Z}$. $\sin(\pi z)$ is entire and $\sin(\pi n) = 0$ for all $n \in \mathbb{Z}$. Thus for prescribed integer values, one may add $\sin(\pi z) \cdot H(z)$ (with H entire) without changing those values.

Removable singularities (Riemann). If f is holomorphic on $0 < |z - a| < r$ and bounded near a , then f extends holomorphically to $z = a$.

How these tools are used in Problem 8.

- (i) $g(z) = f(z) - z$ entire with $g(1/n) = 0$ and $1/n \rightarrow 0 \in \mathbb{C} \Rightarrow g \equiv 0$.
- (ii) Integers have no accumulation in $\mathbb{C}$; non-uniqueness via $\sin(\pi z)$.
- (iii) Zeros at $1/n \rightarrow 0$ in $D(0, 2) \setminus \{-1\}$ force identity with a meromorphic function that has a pole at -1 — contradiction.

Solution to Problem 8.

- (i) Define

$$g(z) := f(z) - z.$$

Then g is entire and $g(1/n) = 0$ for all n . Since $1/n \rightarrow 0$ (an interior point), the identity theorem gives $g \equiv 0$, hence $f(z) \equiv z$.

- (ii) Not necessarily. For example,

$$f(z) = z^2 + \sin(\pi z)$$

is entire and satisfies $f(n) = n^2$ for all $n \in \mathbb{Z}$, yet $f \not\equiv z^2$.

- (iii) Suppose $f(1/n) = 1/(n+1)$ for all n . Then

$$h(z) := f(z) - \frac{z}{z+1}$$

is holomorphic on $D(0, 2) \setminus \{-1\}$, vanishes at $z = 1/n$ for all n , and the set $\{1/n\}$ accumulates at 0 in that domain. By the identity theorem, $h \equiv 0$ there, i.e. $f(z) = \frac{z}{z+1}$

on $D(0, 2) \setminus \{-1\}$. But $\frac{z}{z+1}$ has a pole at $z = -1$, whereas f is holomorphic there — a contradiction. Hence some n satisfies $f(1/n) \neq 1/(n+1)$.

Problem 9 (Casorati–Weierstrass; restated). If f is holomorphic on $D(a, R) \setminus \{a\}$ with an essential singularity at a , then for any $b \in \mathbb{C}$ there exists a sequence $z_n \rightarrow a$ such that $f(z_n) \rightarrow b$.

Solution to Problem 9. If f omitted b near a , then $g = 1/(f - b)$ would be bounded near a , hence removable there by Riemann’s theorem. Thus $f = b + 1/g$ would have at worst a pole/removable singularity at a , contradicting essentiality. *Example.* For $f(z) = e^{1/z}$, $a = 0$, $b = 2$, take

$$z_n = \frac{1}{\operatorname{Log} 2 + 2\pi i n} \Rightarrow e^{1/z_n} = 2.$$

Problem 10 (restated). Let $D \subset \mathbb{C}$ be simply connected and $0 \notin D$. Show that there exists a holomorphic function F on D with $F'(z) = 1/z$ and $e^{F(z)} = z$ for $z \in D$ (i.e. F is a branch of the logarithm on D).

Solution to Problem 10. Fix $z_0 \in D$ and define

$$F(z) = \int_{z_0}^z \frac{d\zeta}{\zeta}.$$

Since $1/\zeta$ is holomorphic on D and D is simply connected, the integral is path-independent; F is holomorphic with $F'(z) = 1/z$. Let $G(z) = e^{F(z)}/z$. Then $G'(z) = 0$, so G is constant. Choosing $F(z_0)$ so that $G \equiv 1$ yields $e^{F(z)} = z$ on D .

Problem 11 (restated). Consider the lacunary series

$$f(z) = \sum_{n=1}^{\infty} z^{2^n}.$$

Show that it defines an analytic function on $D(0, 1)$ but admits no analytic continuation across any point of $|z| = 1$.

Solution to Problem 11. The series converges absolutely and uniformly on compact subsets of $D(0, 1)$, so f is analytic there. Moreover, for $m \geq 1$,

$$(1 - z^{2^m})f(z) = \sum_{j=1}^{m-1} z^{2^j},$$

so each 2^m -th root of unity is a singularity of f (unless trivially canceled). As the union $\bigcup_{m \geq 1} \{\zeta : \zeta^{2^m} = 1\}$ is dense on $|z| = 1$, $|z| = 1$ is a *natural boundary*: no analytic continuation is possible across any boundary point. $\square$

Background for Problems 9–11. *Domains and basic classes.* A *domain* is a nonempty, open, connected subset of $\mathbb{C}$. A function is *holomorphic* on U if it is complex differentiable at every point of U . It is *entire* if holomorphic on $\mathbb{C}$. It is *meromorphic* on U if holomorphic on U except at isolated poles.

Isolated singularities. Let f be holomorphic on $0 < |z - a| < r$. Then a is an *isolated singularity* and exactly one holds:

- **Removable:** f is bounded near $a \Rightarrow f$ extends holomorphically to a (Riemann).
- **Pole:** $|f(z)| \rightarrow \infty$ as $z \rightarrow a$; equivalently, $(z - a)^m f(z)$ extends holomorphically with nonzero value at a for some $m \in \mathbb{N}$.
- **Essential:** neither removable nor pole. The Laurent expansion at a then has infinitely many negative powers.

Casorati–Weierstrass & Picard (context for Problem 9). If a is an essential singularity of f , then the image of any punctured neighborhood of a is dense in $\mathbb{C}$ (Casorati–Weierstrass). A stronger statement (Great Picard) says that near an essential singularity f attains every complex value, with at most one exception, infinitely often.

Identity theorem & isolated zeros. If f, g are holomorphic on a domain U and $f = g$ on a set with a limit point in U , then $f \equiv g$ on U . Consequently, a nonzero holomorphic function on U has zeros that do not accumulate in U .

Path independence and primitives (for Problem 10). If $\omega(z) dz$ is holomorphic on a simply connected domain D , then

$$F(z) := \int_{z_0}^z \omega(\zeta) d\zeta$$

is well defined (independent of path), holomorphic on D , and satisfies $F' = \omega$. In particular, if $0 \notin D$, the 1-form $d\zeta/\zeta$ has the primitive $F(z) = \int_{z_0}^z \frac{d\zeta}{\zeta}$ on D , so e^F gives a single-valued branch of the logarithm.

Winding number criterion for a logarithm. A holomorphic, nowhere-vanishing function g on a domain D admits a holomorphic branch of $\text{Log } g$ on D iff $\int_{\gamma} \frac{g'}{g} dz = 0$ for every closed curve $\gamma \subset D$; equivalently, $g(\gamma)$ has winding number 0 about 0 for all such γ . When D is simply connected and g never vanishes, a branch of $\text{Log } g$ always exists.

Power series, radius of convergence, and natural boundaries (for Problem 11). A power series $\sum_{n \geq 0} a_n z^n$ has a radius of convergence $R \in [0, \infty]$ determined by the Cauchy–Hadamard formula. It defines an analytic function on $|z| < R$ and cannot be analytically continued across points where the function has a singularity. A Jordan curve Γ is a *natural boundary* for f if no analytic continuation of f exists across *any* point of Γ .

Hadamard gap theorem (lacunary series). If $f(z) = \sum_{k \geq 1} a_k z^{n_k}$ with strictly increasing exponents (n_k) satisfying $\liminf_{k \rightarrow \infty} \frac{n_{k+1}}{n_k} > 1$ (“Hadamard gaps”), then the circle of convergence $|z| = R$ is a *natural boundary* for f (unless f is a polynomial). In particular, for $n_k = 2^k$ (ratio = 2) the unit circle is a natural boundary when $R = 1$.

Dense boundary singularities via functional identities (alternate route for Problem 11). If a function f on $|z| < 1$ satisfies identities of the form

$$(1 - z^N) f(z) = P_N(z) \quad (N \in \mathbb{N}),$$

with P_N polynomial and $N \rightarrow \infty$, then every N -th root of unity is a singularity (unless canceled), and the union of these roots for $N \rightarrow \infty$ is dense on $|z| = 1$. Hence $|z| = 1$ is a natural boundary (no analytic continuation across any boundary point).

Problem. Let f be holomorphic on $D(a, R)$. If w satisfies $|w - a| < r < R$, show

$$f^{(n)}(w) = \frac{n!}{2\pi i} \int_{|z-a|=r} \frac{f(z)}{(z-w)^{n+1}} dz.$$

Background. *Residue theorem.* If g is meromorphic on and inside a positively oriented simple closed curve γ (no poles on γ),

$$\int_{\gamma} g(z) dz = 2\pi i \sum \text{Res}(g; a_k).$$

Higher-order pole. If g has a pole of order m at w ,

$$\text{Res}(g; w) = \frac{1}{(m-1)!} \lim_{z \rightarrow w} \frac{d^{m-1}}{dz^{m-1}} \left((z-w)^m g(z) \right).$$

Taylor series. $f(z) = \sum_{k \geq 0} \frac{f^{(k)}(w)}{k!} (z-w)^k$ near w .

Solution (via residues). Let $g(z) = \frac{f(z)}{(z-w)^{n+1}}$. Then g is meromorphic on $|z-a| \leq r$ with a single pole of order $n+1$ at $z=w$. Hence

$$\text{Res}(g; w) = \frac{1}{n!} f^{(n)}(w).$$

Applying the residue theorem to the circle $|z-a|=r$ gives

$$\int_{|z-a|=r} \frac{f(z)}{(z-w)^{n+1}} dz = 2\pi i \text{Res}(g; w) = 2\pi i \frac{f^{(n)}(w)}{n!},$$

which rearranges to the stated formula. For $n=0$ this is the usual Cauchy integral formula.

Examples.

- $f(z) = e^z$: $\frac{n!}{2\pi i} \int_{|z-a|=r} \frac{e^z}{(z-w)^{n+1}} dz = e^w$.
- $f(z) = \sum_{k=0}^m a_k z^k$: the integral vanishes for $n > m$, agreeing with $f^{(n)} \equiv 0$.
- The integral is independent of r as long as $|w-a| < r < R$ (no singularities crossed when deforming the contour).

□

Pole of order 3: coefficients c_{-3}, c_{-2}, c_{-1} and the residue. Suppose f has a pole of order 3 at a . Define

$$h(z) := (z-a)^3 f(z) = \sum_{n=0}^{\infty} d_n (z-a)^n, \quad d_n = \frac{h^{(n)}(a)}{n!}.$$

Then

$$f(z) = \frac{d_0}{(z-a)^3} + \frac{d_1}{(z-a)^2} + \frac{d_2}{(z-a)} + \cdots,$$

so

$$\boxed{c_{-3} = d_0, \quad c_{-2} = d_1, \quad c_{-1} = d_2 = \text{Res}(f; a)}.$$

Example. For $f(z) = \frac{e^z}{(z-a)^3}$ we have $h(z) = e^z$, hence

$$c_{-3} = e^a, \quad c_{-2} = e^a, \quad c_{-1} = \frac{e^a}{2}, \quad \text{Res}(f; a) = \frac{e^a}{2}.$$

Equivalently, $\text{Res}(f; a) = \frac{1}{2!} \frac{d^2}{dz^2} ((z-a)^3 f(z)) \Big|_{z=a}$.

Three distinct poles inside a contour: sum of residues. Let γ be a positively oriented simple closed curve and assume f is meromorphic on and inside γ , with distinct poles a, b, c inside and none on γ . Then the residue theorem gives

$$\boxed{\int_{\gamma} f(z) dz = 2\pi i (\text{Res}(f; a) + \text{Res}(f; b) + \text{Res}(f; c)).}$$

Simple-pole residues. If $f = \frac{p}{q}$ with p, q holomorphic and $q'(a) \neq 0$,

$$\text{Res}(f; a) = \frac{p(a)}{q'(a)}.$$

For example, for $f(z) = \frac{1}{(z-a)(z-b)(z-c)}$,

$$\text{Res}(f; a) = \frac{1}{(a-b)(a-c)}, \quad \text{Res}(f; b) = \frac{1}{(b-a)(b-c)}, \quad \text{Res}(f; c) = \frac{1}{(c-a)(c-b)},$$

and hence

$$\int_{\gamma} f(z) dz = 2\pi i \left(\frac{1}{(a-b)(a-c)} + \frac{1}{(b-a)(b-c)} + \frac{1}{(c-a)(c-b)} \right).$$

Residues from Laurent series at each pole. Let f be meromorphic and let a_j be a pole (order m_j). Its Laurent expansion about a_j is

$$f(z) = \sum_{k=1}^{m_j} \frac{c_{-k}^{(j)}}{(z-a_j)^k} + \sum_{k=0}^{\infty} c_k^{(j)} (z-a_j)^k.$$

Then

$$\boxed{\text{Res}(f; a_j) = c_{-1}^{(j)}}.$$

If γ encloses poles $a_1, \dots, a_m$ and no others, the residue theorem gives

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^m \text{Res}(f; a_j).$$

Problem (Cauchy's derivative formula via residues). Let f be holomorphic on $D(a, R)$. If w satisfies $|w-a| < r < R$, show

$$f^{(n)}(w) = \frac{n!}{2\pi i} \int_{|z-a|=r} \frac{f(z)}{(z-w)^{n+1}} dz.$$

Background. *Residue at a higher-order pole.* If g has a pole of order m at w ,

$$\operatorname{Res}(g; w) = \frac{1}{(m-1)!} \lim_{z \rightarrow w} \frac{d^{m-1}}{dz^{m-1}} \left((z-w)^m g(z) \right).$$

Residue theorem. If g is meromorphic on and inside a positively oriented simple closed curve γ (no poles on γ),

$$\int_{\gamma} g(z) dz = 2\pi i \sum \operatorname{Res}(g; a_k).$$

Taylor series at w . For f holomorphic near w ,

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(w)}{k!} (z-w)^k.$$

Solution. Put $g(z) = \frac{f(z)}{(z-w)^{n+1}}$. Then g is meromorphic on $|z-a| \leq r$, with a single pole at $z = w$ of order $n+1$. Hence

$$\operatorname{Res}(g; w) = \frac{1}{n!} f^{(n)}(w).$$

By the residue theorem on $|z-a|=r$,

$$\int_{|z-a|=r} \frac{f(z)}{(z-w)^{n+1}} dz = 2\pi i \operatorname{Res}(g; w) = 2\pi i \frac{f^{(n)}(w)}{n!},$$

which yields the claimed formula. For $n=0$ this reduces to Cauchy's integral formula.

Examples.

- $f(z) = e^z$: $\frac{n!}{2\pi i} \int_{|z-a|=r} \frac{e^z}{(z-w)^{n+1}} dz = e^w$.
- Polynomial P of degree m : if $n > m$ then the integral (and $P^{(n)}(w)$) equals 0.
- Independence of r : the integral is unchanged for any r with $|w-a| < r < R$ (no singularities are crossed when deforming the contour).

□

Problem. Let $g(z) = \frac{p(z)}{q(z)}$ be a rational function with $\deg q \geq \deg p + 2$. Show that the sum of the residues of g at all its finite poles equals 0.

Background (residue at infinity). For a meromorphic f on the Riemann sphere,

$$\operatorname{Res}(f; \infty) := - \sum_{a \in \mathbb{C}} \operatorname{Res}(f; a).$$

Equivalently, with $w = 1/z$ and $F(w) = f(1/w)$,

$$\operatorname{Res}(f; \infty) = - \operatorname{Res} \left(\frac{F(w)}{w^2}, 0 \right), \quad dz = -\frac{dw}{w^2}.$$

If $f(z) = O(1/z^2)$ as $z \rightarrow \infty$, then $F(w)/w^2$ is holomorphic at $w = 0$, hence $\operatorname{Res}(f; \infty) = 0$.

Solution. Since $\deg q \geq \deg p + 2$, we have $g(z) = O(1/z^2)$ as $z \rightarrow \infty$. Therefore $\text{Res}(g; \infty) = 0$. By the global residue identity,

$$0 = \text{Res}(g; \infty) = - \sum_{a \in \mathbb{C}} \text{Res}(g; a),$$

so $\sum_{a \in \mathbb{C}} \text{Res}(g; a) = 0$, i.e. the sum of the residues at all finite poles is 0.

Checks.

- $g(z) = \frac{1}{z^2 + 1}$: $\text{Res}(g; i) = \frac{1}{2i}$, $\text{Res}(g; -i) = -\frac{1}{2i}$, sum 0.
- $g(z) = \frac{z}{(z^2 + 1)^2}$: residues at i and $-i$ are opposite; the sum is 0.

□

Problem 3. Evaluate the following integrals. (a) $\int_0^\pi \frac{d\theta}{4 + \sin^2 \theta}$.

Using $\int_0^\pi \frac{d\theta}{a + b \sin^2 \theta} = \frac{\pi}{\sqrt{a(a+b)}}$ for $a > 0$, with $a = 4$, $b = 1$,

$$\int_0^\pi \frac{d\theta}{4 + \sin^2 \theta} = \frac{\pi}{\sqrt{4 \cdot 5}} = \boxed{\frac{\pi}{2\sqrt{5}}}.$$

(b) $\int_0^\infty \sin(x^2) dx$.

By the Fresnel/Gaussian argument,

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \cos(x^2) dx = \boxed{\frac{\sqrt{2\pi}}{4} = \frac{\sqrt{\pi}}{2\sqrt{2}}}.$$

(c) $\int_0^\infty \frac{x^2}{(x^2 + 4)^2(x^2 + 9)} dx$.

Let $f(z) = \frac{z^2}{(z^2 + 4)^2(z^2 + 9)}$. The integrand is even; hence

$$\int_0^\infty f(x) dx = \frac{1}{2} \int_{-\infty}^\infty f(x) dx.$$

Poles in the upper half-plane: $2i$ (double), $3i$ (simple).

$$\text{Res}(f; 2i) = -\frac{13i}{200}, \quad \text{Res}(f; 3i) = \frac{3i}{50}, \quad \text{Res}(f; 2i) + \text{Res}(f; 3i) = -\frac{i}{200}.$$

Thus

$$\int_{-\infty}^\infty f(x) dx = 2\pi i \left(-\frac{i}{200} \right) = \frac{\pi}{100}, \quad \int_0^\infty f(x) dx = \boxed{\frac{\pi}{200}}.$$

(d) $\int_0^\infty \frac{\ln(x^2 + 1)}{x^2 + 1} dx$.

With $x = \tan t$ ($t \in [0, \pi/2)$), $x^2 + 1 = \sec^2 t$, $dx = \sec^2 t \, dt$,

$$\int_0^\infty \frac{\ln(x^2 + 1)}{x^2 + 1} dx = \int_0^{\pi/2} \ln(\sec^2 t) dt = 2 \int_0^{\pi/2} \ln(\sec t) dt = -2 \int_0^{\pi/2} \ln(\cos t) dt = \boxed{\pi \ln 2}.$$

□

Problem 4 (restated). For $\alpha \in (-1, 1)$ with $\alpha \neq 0$, compute

$$\int_0^\infty \frac{x^\alpha}{x^2 + x + 1} dx.$$

4. Parameter integral. For $\alpha \in (-1, 1)$, $\alpha \neq 0$,

$$I(\alpha) = \int_0^\infty \frac{x^\alpha}{x^2 + x + 1} dx = \frac{2\pi}{\sqrt{3}} \frac{\sin((\alpha + 1)\pi/3)}{\sin(\pi\alpha)}.$$

Reason. Using the Mellin-type formula $\int_0^\infty \frac{x^{\mu-1}}{x^2 + 2x \cos \phi + 1} dx = \frac{\pi \sin(\mu\phi)}{\sin(\pi\mu) \sin \phi}$ (keyhole contour), with $\mu = \alpha + 1$ and $\phi = \pi/3$.

Problem 5 (restated). Let $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0$ be a polynomial of degree n and set $A = \max\{|a_0|, \dots, |a_{n-1}|\}$. Prove that p has n zeros (counting multiplicity) in the disk $\{|z| < A + 1\}$.

5. Roots inside $|z| < A + 1$. Let $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$ and $A = \max_k |a_k|$. For $|z| \geq A + 1$,

$$|p(z)| \geq |z|^n - \sum_{k=0}^{n-1} |a_k| |z|^k \geq |z|^n - A \frac{|z|^n - 1}{|z| - 1} > 0.$$

Hence p has no zeros for $|z| \geq A + 1$, so all (exactly n) zeros lie in $|z| < A + 1$.

Problem 6 (restated). Let $p(z) = z^5 + z$.

- Find all z with $|z| = 1$ such that $\Im p(z) = 0$.
- Compute $\Re p(z)$ at the points from (a).
- Sketch the curve $p \circ \gamma$ where $\gamma(t) = e^{2\pi it}$.
- Using your sketch, determine for each real x the number of solutions z (counted with multiplicity) to $p(z) = x$ with $|z| < 1$.

6. The map $p(z) = z^5 + z$ on the unit circle. Let $z = e^{i\theta}$.

$$p(e^{i\theta}) = e^{5i\theta} + e^{i\theta} = 2 \cos(2\theta) e^{i3\theta}.$$

(a) *Real axis crossings.* $\Im p = 0 \iff \sin(5\theta) + \sin \theta = 2 \sin(3\theta) \cos(2\theta) = 0$, so $\theta = \frac{m\pi}{3}$ ($m = 0, \dots, 5$) or $\theta = \frac{\pi}{4} + \frac{k\pi}{2}$ ($k = 0, 1, 2, 3$).

(b) *Real parts at these points.* $\Re p = \cos(5\theta) + \cos \theta = 2 \cos(3\theta) \cos(2\theta)$. For $\cos(2\theta) = 0$, $\Re p = 0$; for $\theta = \frac{m\pi}{3}$ the values are 2, 1, -1, -2, -1, 1.

(c) *Sketch.* The curve $p(\gamma)$ ($\gamma(t) = e^{2\pi it}$) is a three-petal curve of maximal radius 2, meeting the real axis at -2, -1, 0, 1, 2.

(d) *Number of solutions to $z^5 + z = x$ with $|z| < 1$, $x \in \mathbb{R}$.* By the argument principle, the number equals the winding number of $p(\gamma)$ about x . From the sketch,

$$N(x) = \begin{cases} 0, & |x| > 2, \\ 1, & 1 < |x| < 2, \\ 2, & 0 < |x| < 1, \\ 1, & x = 0 \text{ (one interior root; the other four sit on/near the boundary).} \end{cases}$$

□

Background for Problems 4–6. *Mellin/keyhole integrals.* For $0 < \Re \mu < 2$ and $0 < \phi < \pi$,

$$\int_0^\infty \frac{x^{\mu-1}}{x^2 + 2x \cos \phi + 1} dx = \frac{\pi \sin(\mu\phi)}{\sin(\pi\mu) \sin \phi}.$$

Derivation uses a keyhole contour with the branch of Log on $\mathbb{C} \setminus (-\infty, 0]$.

Definitions. Mellin transform: $\mathcal{M}[f](s) = \int_0^\infty x^{s-1} f(x) dx$. Keyhole contour: a contour encircling the positive real axis to capture the $2\pi i$ jump of the logarithm.

Cauchy's root bound (polynomials). For $p(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$, every root satisfies

$$|z| < 1 + \max_{0 \leq k \leq n-1} |a_k| =: A + 1.$$

Idea: on $|z| = R > A$, $|z|^n > \sum_{k \leq n-1} |a_k| |z|^k$, hence $|p(z)| > 0$.

Rouché's theorem (context). If f, g are holomorphic and $|f| > |g|$ on a simple closed curve, then f and $f + g$ have the same number of zeros inside.

Argument principle. If f is meromorphic with no zeros/poles on a positively oriented simple closed curve γ ,

$$N - P = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi} \Delta_\gamma \arg f.$$

For $f(z) = p(z) - x$ (polynomial minus real parameter), $P = 0$, and the integral counts the zeros of $p(z) = x$ inside.

Unit circle parametrization for $p(z) = z^5 + z$. For $z = e^{i\theta}$,

$$p(e^{i\theta}) = e^{i5\theta} + e^{i\theta} = 2 \cos(2\theta) e^{i3\theta}.$$

Thus $p(\partial\mathbb{D})$ is a three-petal curve (max radius 2) with real-axis crossings at $-2, -1, 0, 1, 2$.

Tiny examples. (4) *Mellin use.* With $x^2 + x + 1 = x^2 + 2x \cos(\pi/3) + 1$, set $\mu = \alpha + 1$, $\phi = \pi/3$:

$$\int_0^\infty \frac{x^\alpha}{x^2 + x + 1} dx = \frac{2\pi}{\sqrt{3}} \frac{\sin((\alpha+1)\pi/3)}{\sin(\pi\alpha)} \quad (-1 < \alpha < 1, \alpha \neq 0).$$

(5) *Root bound.* If $p(z) = z^4 - 7z^2 + 10$, then $A = \max\{7, 10\} = 10$, so every root has $|z| < 11$.

(6) *Winding numbers for $z^5 + z = x$.* From $p(e^{i\theta}) = 2\cos(2\theta)e^{i3\theta}$, the number of zeros of $z^5 + z = x$ in $|z| < 1$ (counting multiplicity) is

$$N(x) = \begin{cases} 0, & |x| > 2, \\ 1, & 1 < |x| < 2, \\ 2, & 0 < |x| < 1, \\ 1, & x = 0. \end{cases}$$

□

The keyhole (dogbone) contour: method and examples. *Setup.* For integrals of the form

$$\int_0^\infty x^{\mu-1} F(x) dx,$$

choose the branch $z^{\mu-1} = e^{(\mu-1)\text{Log } z}$ with $\text{Log } z = \log |z| + i \text{Arg } z$ and a branch cut along a ray (typically the positive or negative real axis). The *keyhole contour* $\Gamma_{R,\varepsilon}$ wraps once around this cut:

- upper bank: $x \in [\varepsilon, R]$ along the ray, angle 0,
- large circle C_R : $|z| = R$, counterclockwise,
- lower bank: $x \in [R, \varepsilon]$ just below the ray, angle 2π ,
- small circle c_ε : $|z| = \varepsilon$, counterclockwise.

Along the lower bank, $z^{\mu-1}$ picks up the factor $e^{2\pi i(\mu-1)}$.

Vanishing of circular arcs. If $f(z) = z^{\mu-1}/Q(z)$ and Q is a polynomial, then

$$\int_{C_R} f(z) dz \rightarrow 0 \quad (R \rightarrow \infty) \text{ if } \Re \mu < \deg Q, \quad \int_{c_\varepsilon} f(z) dz \rightarrow 0 \quad (\varepsilon \rightarrow 0) \text{ if } \Re \mu > 0.$$

Hence one typically assumes $0 < \Re \mu < \deg Q$.

Bank contributions and jump factor. Let $I = \int_0^\infty \frac{x^{\mu-1}}{Q(x)} dx$. Upper bank contributes I . Lower bank contributes $-e^{2\pi i(\mu-1)} I$ (reversed orientation). Thus

$$\int_{\Gamma_{R,\varepsilon}} \frac{z^{\mu-1}}{Q(z)} dz \xrightarrow[R \rightarrow \infty]{\varepsilon \rightarrow 0} (1 - e^{2\pi i(\mu-1)}) I = 2i e^{i\pi(\mu-1)} \sin(\pi\mu) I.$$

Residue theorem. If Q has zeros a_k not on the cut, then

$$\int_{\Gamma_{R,\varepsilon}} \frac{z^{\mu-1}}{Q(z)} dz \rightarrow 2\pi i \sum_k \text{Res} \left(\frac{z^{\mu-1}}{Q(z)}; a_k \right).$$

Equate with the bank expression to solve for I .

Example 1 (Euler's Beta integral). For $Q(z) = 1 + z$ and $0 < \Re \mu < 1$,

$$\int_0^\infty \frac{x^{\mu-1}}{1+x} dx = \frac{\pi}{\sin(\pi\mu)}.$$

Example 2 (quadratic denominator). For $Q(z) = z^2 + 2z \cos \phi + 1 = (z - e^{i\phi})(z - e^{-i\phi})$ with $0 < \phi < \pi$ and $0 < \Re \mu < 2$,

$$\int_0^\infty \frac{x^{\mu-1}}{x^2 + 2x \cos \phi + 1} dx = \frac{\pi \sin(\mu\phi)}{\sin(\pi\mu) \sin \phi}.$$

Sketch. The poles are at $e^{\pm i\phi}$; compute

$$\operatorname{Res} \left(\frac{z^{\mu-1}}{(z - e^{i\phi})(z - e^{-i\phi})}; e^{i\phi} \right) = \frac{e^{i\phi(\mu-1)}}{2i \sin \phi}, \quad \operatorname{Res}(\cdots; e^{-i\phi}) = \frac{e^{-i\phi(\mu-1)}}{-2i \sin \phi}.$$

Sum and equate with the bank expression.

Pitfalls. (1) Be consistent with the branch cut; (2) ensure the circular arcs vanish (conditions on $\Re \mu$); (3) avoid poles on the cut (move the cut, or indent and take principal values if needed).

Problem 7 (cotangent trick and square contour). Let $N \in \mathbb{N}$ and let γ_N be the square contour with vertices $(\pm 1 \pm i)(N + \frac{1}{2})$, oriented positively.

(i) Show that there exists a constant $C > 0$, independent of N , such that

$$|\cot(\pi z)| < C \quad \text{for all } z \in \gamma_N.$$

(ii) By integrating $\frac{\pi \cot(\pi z)}{z^2 + 1}$ around γ_N , prove that

$$\sum_{n=0}^{\infty} \frac{1}{n^2 + 1} = \frac{1 + \pi \coth \pi}{2}.$$

(iii) Evaluate the alternating series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2 + 1}.$$

7. (i) Uniform bound for $|\cot \pi z|$ on γ_N . With $z = x + iy$ one has

$$|\sin(\pi z)|^2 = \sin^2(\pi x) + \sinh^2(\pi y), \quad |\cos(\pi z)|^2 = \cos^2(\pi x) + \sinh^2(\pi y).$$

On the *vertical* sides $x = \pm(N + \frac{1}{2})$ we get $\sin(\pi x) = \pm 1$, $\cos(\pi x) = 0$, hence

$$\cot(\pi z) = -i \tanh(\pi y), \quad |\cot(\pi z)| \leq 1.$$

On the *horizontal* sides $y = \pm(N + \frac{1}{2})$,

$$|\cot(\pi z)|^2 = \frac{\cos^2(\pi x) + \sinh^2(\pi y)}{\sin^2(\pi x) + \sinh^2(\pi y)} \leq 1 + \frac{1}{\sinh^2(\pi|y|)} \leq 1 + \frac{1}{\sinh^2(\pi/2)}.$$

Thus there is a constant $C = \max\{1, \sqrt{1 + \sinh^{-2}(\pi/2)}\}$, independent of N , such that $|\cot \pi z| < C$ on γ_N .

(ii) $\sum_{n=0}^{\infty} \frac{1}{n^2 + 1}$. Let $F(z) = \frac{\pi \cot(\pi z)}{z^2 + 1}$. Then on γ_N , $|F(z)| \lesssim C/|z|^2$, so $\int_{\gamma_N} F(z) dz \rightarrow 0$ as $N \rightarrow \infty$. Residues:

$$\operatorname{Res}(F; n) = \frac{1}{n^2 + 1} \quad (n \in \mathbb{Z}), \quad \operatorname{Res}(F; i) = \operatorname{Res}(F; -i) = \frac{\pi \cot(\pi i)}{2i}.$$

Since $\cot(\pi i) = -i \coth \pi$, we have $\operatorname{Res}(F; i) + \operatorname{Res}(F; -i) = -\pi \coth \pi$. Hence, letting $N \rightarrow \infty$,

$$0 = \int_{\gamma_N} F = 2\pi i \left(\sum_{n \in \mathbb{Z}} \frac{1}{n^2 + 1} - \pi \coth \pi \right) \Rightarrow \sum_{n \in \mathbb{Z}} \frac{1}{n^2 + 1} = \pi \coth \pi.$$

By symmetry,

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{n^2 + 1} = \frac{1 + \pi \coth \pi}{2}}.$$

(iii) $\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2 + 1}$. Set $G(z) = \frac{\pi \csc(\pi z)}{z^2 + 1}$. Again $\int_{\gamma_N} G \rightarrow 0$ and

$$\operatorname{Res}(G; n) = \frac{(-1)^n}{n^2 + 1} \quad (n \in \mathbb{Z}), \quad \csc(\pi i) = \frac{1}{\sin(\pi i)} = \frac{1}{i \sinh \pi} = -i \operatorname{csch} \pi.$$

Therefore

$$\operatorname{Res}(G; i) + \operatorname{Res}(G; -i) = \frac{\pi \csc(\pi i)}{2i} + \frac{\pi \csc(-\pi i)}{-2i} = -\pi \operatorname{csch} \pi.$$

Thus

$$\sum_{n \in \mathbb{Z}} \frac{(-1)^n}{n^2 + 1} = \pi \operatorname{csch} \pi, \quad \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2 + 1} = \frac{1 + \pi \operatorname{csch} \pi}{2}}.$$

□

Complex cotangent: identities and geometry. *Core identities.* For $z = x + iy$,

$$\sin(\pi z) = \sin(\pi x) \cosh(\pi y) + i \cos(\pi x) \sinh(\pi y), \quad \cos(\pi z) = \cos(\pi x) \cosh(\pi y) - i \sin(\pi x) \sinh(\pi y),$$

hence

$$|\sin(\pi z)|^2 = \sin^2(\pi x) + \sinh^2(\pi y), \quad |\cos(\pi z)|^2 = \cos^2(\pi x) + \sinh^2(\pi y).$$

Also

$$\cot z = i \coth(iz) = i \frac{e^{2iz} + 1}{e^{2iz} - 1}.$$

Poles, residues, partial fractions. $\cot$ has simple poles at $k\pi$ with residue 1; $\pi \cot(\pi z)$ has simple poles at integers with residue 1.

$$\pi \cot(\pi z) = \sum_{n \in \mathbb{Z}} \frac{1}{z - n}.$$

Images of horizontal lines $y = \text{const.}$ Write

$$\cot(x + iy) = \frac{\sin 2x - i \sinh 2y}{\cosh 2y - \cos 2x} = u(x) + iv(x).$$

Eliminating x yields the circle

$$\boxed{u^2 + (v + \coth 2y)^2 = \text{csch}^2 2y},$$

i.e. for fixed y the image is a circle centered at $-i \coth 2y$ with radius $\text{csch} 2y$ (orthogonal to the imaginary axis). As $y \rightarrow \pm\infty$ this circle collapses to the point $\mp i$; as $y \rightarrow 0$ it expands and approaches the real axis.

Images of vertical lines $x = \text{const.}$ For fixed x , as y varies,

$$\cot(x + iy) \rightarrow -i \ (y \rightarrow +\infty), \quad \cot(x + iy) \rightarrow +i \ (y \rightarrow -\infty),$$

and $\cot(x + 0i) = \cot x \in \mathbb{R}$. Thus the image is a circular arc joining $\pm i$ and crossing the real axis at $\cot x$ (orthogonal to the family in the previous paragraph).

Strips, bands, and Möbius viewpoint. Because $\cot z = \tan(\frac{\pi}{2} - z)$, the vertical strip $\{0 < \Re z < \pi\}$ is mapped bijectively onto $\mathbb{C}$, and periodicity $\cot(z + \pi) = \cot z$ tiles the plane by such strips. Via $e^{2iz} = e^{-2y} e^{2ix}$ a horizontal strip $y \in (y_1, y_2)$ maps to an annulus; the Möbius map $w \mapsto i \frac{w+1}{w-1}$ then sends annuli to circular bands bounded by the circles $u^2 + (v + \coth 2y)^2 = \text{csch}^2 2y$.

Bounds on large squares (for residue summations). On the square with vertices $(\pm 1 \pm i)(N + \frac{1}{2})$,

$$|\cot(\pi z)| \leq 1 \text{ on vertical edges,} \quad |\cot(\pi z)|^2 \leq 1 + \frac{1}{\sinh^2(\pi/2)} \text{ on horizontal edges,}$$

so there is a constant C independent of N with $|\cot(\pi z)| < C$ on the boundary.

Asymptotics and symmetries. $\cot(-z) = -\cot z$, $\overline{\cot(\bar{z})} = \cot z$, and $\cot(x + iy) \rightarrow \mp i$ as $y \rightarrow \pm\infty$. These give quick qualitative sketches of level sets and images.

Short derivation of the horizontal-circle equation. Let $z = x + iy$ and set

$$\cot z = \frac{\sin 2x - i \sinh 2y}{\cosh 2y - \cos 2x} = u + iv, \quad D := \cosh 2y - \cos 2x, \quad S := \sinh 2y, \quad C := \cosh 2y.$$

Then

$$u = \frac{\sin 2x}{D}, \quad v = -\frac{S}{D} \implies D = -\frac{S}{v}.$$

Hence

$$\sin 2x = uD = -\frac{uS}{v}, \quad \cos 2x = C - D = C + \frac{S}{v}.$$

Now

$$u^2 + \left(v + \frac{C}{S}\right)^2 = \frac{1}{D^2} \left[\sin^2 2x + \frac{(1 - C \cos 2x)^2}{S^2} \right].$$

Using $D = C - \cos 2x$ and $\cosh^2 2y - \sinh^2 2y = 1$ we get

$$S^2 \sin^2 2x + (1 - C \cos 2x)^2 = (C - \cos 2x)^2 = D^2.$$

Therefore

$$\boxed{u^2 + (v + \coth 2y)^2 = \text{csch}^2 2y},$$

i.e. for fixed y the image is a circle centered at $-i \coth(2y)$ with radius $\text{csch}(2y)$.

Hyperbolic cosecant.

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad \operatorname{sech} x = \frac{1}{\cosh x}, \quad \coth x = \frac{\cosh x}{\sinh x}.$$

Parity and derivatives:

$$\operatorname{csch}(-x) = -\operatorname{csch} x, \quad \frac{d}{dx} \operatorname{csch} x = -\operatorname{csch} x \coth x, \quad \frac{d}{dx} \coth x = -\operatorname{csch}^2 x.$$

Identity and asymptotics:

$$\coth^2 x - \operatorname{csch}^2 x = 1, \quad \operatorname{csch} x \sim 2e^{-|x|} \quad (|x| \rightarrow \infty).$$

Background and geometry for $\sin z$. *Core identities.* For $z = x + iy$,

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad |\sin z|^2 = \sin^2 x + \sinh^2 y.$$

Symmetries: $\overline{\sin z} = \sin \bar{z}$, $\sin(-z) = -\sin z$, $\sin(z + 2\pi) = \sin z$. Zeros are simple at $z = n\pi$ ($n \in \mathbb{Z}$) since $\sin'(n\pi) = \cos(n\pi) \neq 0$.

Images of horizontal lines $y = \text{const.}$ Let $w = \sin z = u + iv$. For fixed y ,

$$u = \sin x \cosh y, \quad v = \cos x \sinh y.$$

Eliminating x gives the ellipse

$$\left(\frac{u}{\cosh y} \right)^2 + \left(\frac{v}{\sinh y} \right)^2 = 1.$$

Thus the image is an ellipse centered at 0 with semiaxes $\cosh y$ (real axis) and $\sinh y$ (imaginary axis). Since $\cosh^2 y - \sinh^2 y = 1$, the foci lie at ± 1 , and

$$|w - 1| + |w + 1| = 2 \cosh y.$$

Images of vertical lines $x = \text{const.}$ For fixed x ,

$$u = \sin x \cosh y, \quad v = \cos x \sinh y$$

and eliminating y using $\cosh^2 y - \sinh^2 y = 1$ yields

$$\left(\frac{u}{\sin x} \right)^2 - \left(\frac{v}{\cos x} \right)^2 = 1 \quad (\sin x \cos x \neq 0).$$

Hence the image is a hyperbola with the same foci ± 1 . Degenerate cases: $x = n\pi$ gives $u = 0$ (imaginary axis) and $x = \frac{\pi}{2} + n\pi$ gives $v = 0$ (real axis).

Strips and periodicity. One may take the fundamental strip $S = \{0 < \Re z < \pi\}$. On $x = 0$ or $x = \pi$ the image is purely imaginary; in the interior the images of horizontal/vertical lines form the orthogonal confocal families of ellipses and hyperbolas with foci ± 1 . (Branches of $\arcsin$ use the slits $(-\infty, -1] \cup [1, \infty)$.)

Background and geometry for $\cos z$. *Core identities.* For $z = x + iy$,

$$\cos z = \cos x \cosh y - i \sin x \sinh y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y.$$

Symmetries: $\overline{\cos z} = \cos \bar{z}$, $\cos(-z) = \cos z$, $\cos(z + 2\pi) = \cos z$. Zeros are simple at $z = \frac{\pi}{2} + n\pi$ since $\cos'(z) = -\sin z$.

Images of horizontal lines $y = \text{const.}$ Let $w = \cos z = u + iv$. For fixed y ,

$$u = \cos x \cosh y, \quad v = -\sin x \sinh y.$$

Eliminating x gives the ellipse

$$\left(\frac{u}{\cosh y} \right)^2 + \left(\frac{v}{\sinh y} \right)^2 = 1.$$

Thus the image is an ellipse centered at 0 with semiaxes $\cosh y$ (real axis) and $\sinh y$ (imaginary axis); since $\cosh^2 y - \sinh^2 y = 1$, the foci are ± 1 and

$$|w - 1| + |w + 1| = 2 \cosh y.$$

Images of vertical lines $x = \text{const.}$ For fixed x ,

$$u = \cos x \cosh y, \quad v = -\sin x \sinh y.$$

Eliminating y using $\cosh^2 y - \sinh^2 y = 1$ yields

$$\left(\frac{u}{\cos x} \right)^2 - \left(\frac{v}{\sin x} \right)^2 = 1 \quad (\sin x \cos x \neq 0),$$

a hyperbola with the same foci ± 1 . Degenerate cases: $x = n\pi$ gives $v = 0$ (real axis) and $x = \frac{\pi}{2} + n\pi$ gives $u = 0$ (imaginary axis).

Strips and periodicity. A fundamental strip is $S = \{0 < \Re z < \pi\}$. On $x = 0$ or $x = \pi$ the image is real; inside S the images of horizontal/vertical lines form orthogonal confocal ellipses and hyperbolas with foci ± 1 . (Branches of $\arccos$ use the slits $(-\infty, -1] \cup [1, \infty)$.)

Background and geometry for $\tan z$. *Core identities.* For $z = x + iy$,

$$\tan z = \frac{\sin z}{\cos z} = \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y}.$$

Thus, writing $w = \tan z = u + iv$,

$$u = \frac{\sin 2x}{\cos 2x + \cosh 2y}, \quad v = \frac{\sinh 2y}{\cos 2x + \cosh 2y}.$$

Zeros, poles, residues. $\tan z$ has simple zeros at $z = n\pi$ and simple poles at $z = \frac{\pi}{2} + n\pi$ ($n \in \mathbb{Z}$). At each pole, $\text{Res}(\tan; z_0) = 1$. The partial fraction expansion is

$$\pi \tan(\pi z) = \sum_{n \in \mathbb{Z}} \frac{1}{z - (n + \frac{1}{2})}.$$

Symmetries: $\tan(-z) = -\tan z$, $\overline{\tan z} = \tan \bar{z}$, and $\tan(z + \pi) = \tan z$. As $|y| \rightarrow \infty$, $\tan(x + iy) \rightarrow i \operatorname{sgn}(y)$.

Images of horizontal lines $y = \text{const.}$ Fix y and set $C = \cosh(2y)$, $S = \sinh(2y)$, $D = \cos(2x) + C$. Then

$$u = \frac{\sin 2x}{D}, \quad v = \frac{S}{D}.$$

Eliminating x (using $D = S/v$) yields the circle

$$\boxed{u^2 + \left(v - \frac{\cosh(2y)}{\sinh(2y)}\right)^2 = \frac{1}{\sinh^2(2y)}}.$$

Hence the image is a circle centered at $-i \frac{\cosh(2y)}{\sinh(2y)}$ on the imaginary axis (note the sign in $u + iv$) with radius $\frac{1}{|\sinh(2y)|}$. As $y \rightarrow \pm\infty$ the circle shrinks to $\pm i$; as $y \rightarrow 0$ it expands toward the real axis.

Images of vertical lines $x = \text{const.}$ For fixed x , as y varies,

$$\tan(x + iy) \rightarrow i \ (y \rightarrow +\infty), \quad \tan(x + iy) \rightarrow -i \ (y \rightarrow -\infty),$$

and $\tan(x + 0i) = \tan x \in \mathbb{R}$. Thus the image is a circular arc joining $\pm i$ and crossing the real axis at $\tan x$ (orthogonal to the family above).

Strips and periodicity. The vertical strip $S = \{-\frac{\pi}{2} < \Re z < \frac{\pi}{2}\}$ is mapped bijectively onto $\mathbb{C}$ by $\tan$; its boundary lines map to ∞ (simple poles). Periodicity $\tan(z + \pi) = \tan z$ tiles the plane by translates of S .

Background and geometry for $\cot z$

Core identities (for $z = x + iy$)

$$\sin(\pi z) = \sin(\pi x) \cosh(\pi y) + i \cos(\pi x) \sinh(\pi y), \quad \cos(\pi z) = \cos(\pi x) \cosh(\pi y) - i \sin(\pi x) \sinh(\pi y).$$

Hence

$$|\sin(\pi z)|^2 = \sin^2(\pi x) + \sinh^2(\pi y), \quad |\cos(\pi z)|^2 = \cos^2(\pi x) + \sinh^2(\pi y), \quad \cot(\pi z) = \frac{\cos(\pi z)}{\sin(\pi z)}.$$

Symmetries: $\overline{\cot(\bar{z})} = \cot z$, $\cot(-z) = -\cot z$, $\cot(z + \pi) = \cot z$.

Zeros, poles, residues

$\cot(\pi z)$ has simple poles at $z \in \mathbb{Z}$, each with residue $1/\pi$. Equivalently, $\pi \cot(\pi z)$ has residue 1 at every integer and no other singularities.

Partial fractions (two precise forms)

(Mittag-Leffler; converges locally uniformly on $\mathbb{C} \setminus \mathbb{Z}$)

$$\boxed{\pi \cot(\pi z) = \frac{1}{z} + \sum_{n \neq 0} \left(\frac{1}{z - n} + \frac{1}{n} \right)}.$$

(Symmetric / principal value)

$$\boxed{\pi \cot(\pi z) = \lim_{N \rightarrow \infty} \sum_{n=-N}^N \frac{1}{z - n}}.$$

Uniform bound on large squares (for residue sums)

On the vertical sides $x = N + \frac{1}{2}$,

$$\cot(\pi z) = \frac{\cos(\pi x) \cosh(\pi y) - i \sin(\pi x) \sinh(\pi y)}{\sin(\pi x) \cosh(\pi y) + i \cos(\pi x) \sinh(\pi y)} = -i \tanh(\pi y) \Rightarrow |\cot(\pi z)| \leq 1.$$

On the horizontal sides $y = \pm(N + \frac{1}{2})$,

$$|\cot(\pi z)|^2 = \frac{\cos^2(\pi x) + \sinh^2(\pi y)}{\sin^2(\pi x) + \sinh^2(\pi y)} \leq 1 + \frac{1}{\sinh^2(\pi|y|)} \leq 1 + \frac{1}{\sinh^2(\pi/2)}.$$

Thus a single constant $C > 0$ works for all N : $|\cot(\pi z)| < C$ on the square.

Images of horizontal lines $y = \text{const}$ (circles)

Using double angles,

$$\cot z = \frac{\cos z}{\sin z} = \frac{\sin 2x - i \sinh 2y}{\cosh 2y - \cos 2x} = u + iv.$$

Let $C := \cosh(2y)$, $S := \sinh(2y)$, $D := C - \cos 2x$. Then

$$u = \frac{\sin 2x}{D}, \quad v = -\frac{S}{D} \Rightarrow D = -\frac{S}{v}.$$

So $\sin 2x = uD = -\frac{uS}{v}$ and $\cos 2x = C - D = C + \frac{S}{v}$. Using $\sin^2 2x + \cos^2 2x = 1$ and $C^2 - S^2 = 1$ we obtain

$$\boxed{u^2 + \left(v + \frac{C}{S}\right)^2 = \frac{1}{S^2}} \iff \boxed{u^2 + \left(v + \frac{\cosh(2y)}{\sinh(2y)}\right)^2 = \frac{1}{\sinh^2(2y)}}.$$

Thus the image is a circle centered at $-i \frac{\cosh(2y)}{\sinh(2y)}$ with radius $\frac{1}{|\sinh(2y)|}$. As $y \rightarrow +\infty$, $\cot(x + iy) \rightarrow -i$; as $y \rightarrow -\infty$, $\cot(x + iy) \rightarrow +i$.

Images of vertical lines $x = \text{const}$ (circular arcs)

For fixed x ,

$$\cot(x + iy) \rightarrow -i \quad (y \rightarrow +\infty), \quad \cot(x + iy) \rightarrow +i \quad (y \rightarrow -\infty), \quad \cot(x + 0) = \cot x \in \mathbb{R}.$$

Thus each vertical line maps to a circular arc joining $\pm i$ and crossing the real axis at $\cot x$; these arcs are orthogonal to the circles above.

Strips and periodicity

The vertical strip $S = \{0 < \Re z < \pi\}$ is mapped bijectively onto $\mathbb{C}$ by $\cot$; its boundary lines map to ∞ (simple poles). Periodicity $\cot(z + \pi) = \cot z$ tiles the plane by translates of S .